

Chapter 155 — Prostate Biopsy: Techniques and Imaging

- **Transrectal ultrasonography (TRUS)** is a standard urological tool underpinning initial and active-surveillance biopsy, low- and high-dose-rate brachytherapy, cryotherapy and high-intensity focused ultrasound (HIFU).
- **Radiotherapy:** fiducial and radiofrequency markers for real-time tracking; **polyethylene glycol hydrogel spacer** in anterior perirectal fat to cut rectal dose and toxicity. **Benign:** benign prostatic hyperplasia (BPH) planning, some male infertility. **Commonest use: cancer biopsy.**
- Detection improved with **prostate-specific antigen (PSA) screening** and systematic biopsy; screening remains controversial.
- **Historic gold standard: 12-core TRUS-guided transrectal biopsy under local anaesthesia**, now rapidly challenged by the **TRUS-guided transperineal approach** (Europe, Australia, United States).
- **Historical sequence:** TRUS first described by Watanabe (1968) → digitally directed biopsy, common until the late 1980s → TRUS-directed biopsy → **systematic sextant (six-core) protocol, Hodge (1989)** → recommended cores increased to 12, while PSA screening expanded the biopsied population.
- **Up to 1.3 million biopsies a year in the United States.** With significant and insignificant cancer both prevalent, effort centres on indications, technique and limiting complications.

ULTRASONOGRAPHIC ANATOMY OF THE PROSTATE

- **Between bladder neck and urogenital diaphragm, directly anterior to the rectum:** ideal for transrectal imaging.
- **Pathological zones:** anterior fibromuscular stroma (no glandular tissue), transition zone (TZ), central zone (CZ), periurethral zone and peripheral zone (PZ). **They are not visible sonographically as distinct entities.**
- **What TRUS does separate**, especially in glands with significant BPH:
 - **Posterior CZ and PZ**, where most adenocarcinomas arise: **homogeneous and echogenic.**
 - **Anterior TZ:** **heterogeneous.**
- **Corpora amylacea:** surgical-capsule calcifications outlining the PZ–TZ plane. Diffuse calcification is **normal and age-related**; larger symptomatic calculi may reflect infection or inflammation and need evaluation.
- **Prostatic urethra:** midline, so seen whole only in the **sagittal plane**; hypoechoic lumen, periurethral calcification may outline it. Internal sphincter (bladder neck → verumontanum) forms a **hypoechoic ring, funnelling the proximal urethra**. At the **verumontanum** it angles anteriorly, giving an **anteriorly concave sagittal course**.
- **Seminal vesicles (SVs):** paired, posterior at the base, smooth, saccular and symmetrical. Normal size **4.5–5.5 cm long × 2 cm wide**.
 - **Cystic SV mass: presumptively benign. Solid SV mass: significant probability of malignancy**, especially with a primary neoplasm elsewhere. Consider **schistosomiasis** for a solid mass in endemic areas.
 - **Absent SV → ipsilateral renal agenesis in 79%.**
- **Vasa and ejaculatory ducts:** ampullae run above the ipsilateral SV, dive caudally medial to it and **fuse with it to form the ejaculatory duct**, which enters posteriorly (occasionally hypoechoic) and **empties at the verumontanum**, paralleling the distal urethra.

- **Figure 155.1.** Transverse and sagittal views at the verumontanum labelling anterior fibromuscular stroma, dorsal vein complex, levators, neurovascular bundle, ejaculatory ducts, urethra, bladder and zones.

Concept — Zonal anatomy is **pathological, not sonographic**. TRUS reliably shows only the **posterior echogenic PZ/CZ versus the anterior heterogeneous TZ**, and that is the distinction biopsy needs, because cancer is mostly posterior.

GREY-SCALE TRANSRECTAL ULTRASONOGRAPHY

- **Commonest prostate imaging modality:** cancer detection, infertility, volume for brachytherapy and BPH.
- **Probes:** side-fire, end-fire, biplane; **6–10 MHz**; presets on modern machines. **Biplane** (axial + sagittal linear array) suits brachytherapy, ablation and transperineal work. End- and side-fire scan an arc of nearly **180°**. **End-fire vs side-fire: no difference in detection**, initial or repeat (prospective studies).

PROBE FREQUENCY: RESOLUTION VERSUS DEPTH

	Higher frequency (standard 7 MHz)	Lower frequency (older 4 MHz)
Spatial resolution	High	Lower
Focal range	1–4 cm from transducer: closer	2–8 cm: deeper
Best for	PZ, where most cancers arise	Anterior delineation of large glands, so more accurate volume
Weakness	Poorer depth	Poor internal architecture

- **Higher frequency → focal range closer to the transducer;** modern systems adjust frequency mid-examination to the depth of interest.
- **Coupling medium: ultrasound does not cross air**, so gel sits between transducer and rectum; with a sheath, **both inside and outside the sheath**.

Machine Settings

- **Magnification: low for measurement** (whole gland on one screen); **high for biopsy** (to follow the needle).
- **Gain:** set per patient (sometimes per region) so the **normal PZ is medium grey**, the reference for **hypoechoic** (darker), **isoechoic**, **hyperechoic** (lighter) and **anechoic** (black).

Techniques

- **Complete evaluation:** scan in sagittal and transverse planes to calculate volume → inspect CZ and PZ for hypoechoic lesions and contour abnormalities → assess periprostatic fat planes → visualise SVs and vasa fully.

Probe Manipulation

- **Transrectal: left lateral decubitus. Transperineal: lithotomy.**

PROBE MANIPULATION BY PLANE AND PROBE TYPE

Plane	Probe	Manoeuvre	Result
Transverse	Radial or biplane (dedicated transverse crystal)	Advance or withdraw	Cephalad: base, SVs, bladder neck. Caudal (toward anal sphincter): apex, proximal urethra
Transverse	End- or side-fire	Angle handle , anal sphincter as fulcrum	Handle toward scrotum → cephalad ; toward sacrum → caudal
Sagittal	Side-fire (usually)	Rotation	Clockwise → left side ; anticlockwise → right

Plane	Probe	Manoeuvre	Result
Sagittal	End-fire (usually)	Angle up or down , anal sphincter as fulcrum	In left lateral decubitus: handle down (floor) → right side; handle up (ceiling) → left
Sagittal, lithotomy	Freehand biplane	Partial rotation plus angling	Excellent sagittal views
Sagittal, lithotomy	On a stabiliser (brachytherapy stepper)	Rotation only	No angling possible

- Urologists **prefer angling**: cystoscope-like and less uncomfortable. **Reprocessing**: American Urological Association (AUA)/Society of Urological Nurses and Associates white paper (2012, updated 2016).

Concept — Every transrectal manoeuvre uses the **anal sphincter as a fulcrum**, so the **image moves opposite to the handle**. The one exception is rotation, which has no fulcrum.

Volume Calculations

- **Planes**: transverse and anteroposterior (AP) at the widest axial diameter; **longitudinal sagittally, just off midline** (bladder neck obscures the cephalad extent). **AP is better sagittal** (perpendicular to longitudinal); **axial AP overestimates** if the plane is oblique.

PROSTATE VOLUME FORMULAS

Assumed shape	Formula	Dimensions needed
Ellipsoid	$\pi/6 \times \text{transverse} \times \text{AP} \times \text{longitudinal}$	3
Sphere	$\pi/6 \times \text{transverse}^3$	1
Prolate (egg-shaped) spheroid	$\pi/6 \times \text{transverse}^2 \times \text{AP}$	2
Planimetry (no shape assumed)	Sum of serial transverse areas	Serial images; most precise

- Despite the geometric assumptions, **all formulas correlate >0.90 with radical prostatectomy specimen weight**, because $1 \text{ cm}^3 \approx 1 \text{ g}$ of prostate.
- **Normal size**: mature gland **20–25 g**, stable until about **age 50**; average at **60–70 years** $\approx 48 \text{ g}$.
- **AUA BPH/lower urinary tract symptoms (LUTS) guideline (2021)**: assess size (TRUS, abdominal ultrasound, cystoscopy or cross-sectional imaging) **before intervention**, as drugs and operations suit specific size ranges; size also shapes cancer surgery.
- **Planimetry**, used when precision matters (brachytherapy planning): lithotomy → probe mounted on a stepper → serial transverse images at set intervals (**3–5 mm**) through the gland → area of each → sum and integrate to give volume.
- **PSA density (PSAD) = serum PSA ÷ volume**: separates benign from malignant causes of raised PSA with a benign digital rectal examination (DRE). **>0.15 ng/mL/cc suggests malignancy**. For repeat-biopsy cancer: **moderate sensitivity, poor specificity; high operator variability**. Free-to-total PSA, other biomarkers and multiparametric MRI (mpMRI) now give similar information.

Cystic Lesions of the Prostate

- **Common on TRUS**. Simple cysts look the same as elsewhere in the body: **thin-walled, anechoic, with posterior acoustic enhancement**. Congenital or acquired; **rarely clinically significant**.

PROSTATIC CYSTS: MÜLLERIAN VERSUS WOLFFIAN VERSUS ACQUIRED

Cyst	Origin	Position and appearance	Associations
Enlarged prostatic utricle	Müllerian	Midline anechoic diverticulum of the posterior urethra at the verumontanum	Genital anomalies: hypospadias (commonest) , ambiguous genitalia, undescended testes, congenital urethral polyps

Cyst	Origin	Position and appearance	Associations
Müllerian duct cyst	Müllerian (ducts fail to fuse with urethra)	Midline , ovoid to pear-shaped, neck toward the verumontanum	Screen for unilateral renal agenesis
Ejaculatory duct cyst	Wolffian	Small, off the midline	Ejaculatory duct obstruction; azoospermia
SV cyst	Wolffian (congenital or acquired ejaculatory duct obstruction)	Lateral, paraprostatic	Cystic renal disease; renal agenesis in up to two-thirds
Acquired TZ cyst	Haemorrhagic degeneration of BPH nodules	Within TZ	None
Acquired outer-gland cyst	No proven cause	Outer gland	None

- **Müllerian = midline; wolffian = lateral** (ejaculatory duct cyst just off midline). **Renal agenesis is far commoner with wolffian SV cysts**, though both warrant renal assessment.

Prostate Cancer Imaging on Transrectal Ultrasonography

- Once thought pathognomonic; now **sample every hypoechoic PZ lesion**, but **absence does not preclude biopsy: 39% of cancers are isoechoic, up to 1% hyperechoic**.
- **Split evidence**: one large series found **equal detection with or without a lesion**; another found **lesion cores almost twice as often positive**, so targeting remains worthwhile.
- **False-positive hypoechoic lesions**: commonest are **BPH foci and acute or chronic prostatitis**; also granulomatous prostatitis, prostatic infarct and lymphoma.
- Note **contour abnormalities** and **PZ asymmetry**. **Extraprostatic extension (EPE): focal loss of bright periprostatic fat**; microfoci invisible.

KEY POINTS: GREY-SCALE TRANSRECTAL ULTRASONOGRAPHY

- Zonal anatomy is not visible, but **PZ is distinguishable from TZ**, so biopsies can be aimed at the PZ.
- Benign cysts: **thin-walled with through-transmission**; may become symptomatic if infected.
- **Planimetry with a template** is needed for highly accurate volume (brachytherapy).
- Sample hypoechoic foci, but **up to 39% of cancers are invisible** on grey scale.

Transrectal Ultrasonography Appearance After Treatment

TRUS APPEARANCE AFTER TREATMENT

Treatment	Volume change	Appearance and interpretation
External-beam radiotherapy (monotherapy)	Decreased by 6 months	Diffusely hypoechoic , poorly defined anatomy. Large non-responding tumours barely change; small responding foci become isoechoic . TRUS correlates poorly with pathology and outcome
Permanent seed brachytherapy	Initial oedema, then shrinkage: about one-third at 1 year, over half by 8 years . Unaffected by neoadjuvant hormones	Seeds hyperechoic with posterior shadowing . Ideal implant: even distribution, periurethral sparing , no errant extraprostatic seeds

Treatment	Volume change	Appearance and interpretation
Androgen deprivation (luteinising hormone-releasing hormone analogues)	Average ≈30% decrease : up to 60% in large , as little as 10% in small glands	Occurs with or without cancer
5α-reductase inhibitor (finasteride)	≈21% at 6 months	None specific
Radical prostatectomy	Not applicable	Normal: smooth tapering of bladder neck to urethra . Nodule anterior to anastomosis = ligated dorsal vein complex . Nodular bladder neck may reflect the tapering technique (fish-mouth, tennis-racket, parachuting)

- **Suspicious after prostatectomy**: any hyper- or hypoechoic lesion or **broken retroanastomotic fat plane**; recurrence is **hypoechoic in 75–95%**; colour Doppler helps.
- **Blind anastomotic biopsy for PSA-only recurrence (no nodule) is usually uninformative**; biopsy of a **TRUS-visible lesion, even with normal DRE, can be diagnostic**. Contrast pelvic MRI or prostate-specific membrane antigen (PSMA) positron emission tomography (PET) with CT or MRI now finds targets.

Concept — Every treatment *shrinks the gland and flattens its echotexture toward hypoechoic*, erasing the grey-scale contrast TRUS relies on. After treatment, grey-scale TRUS is therefore a **targeting aid for visible abnormalities, not a staging tool**.

Transrectal Ultrasonography and Other Malignancies

OTHER MALIGNANCIES ON TRUS

Tumour	TRUS appearance	Pitfall or note
Transitional cell (urothelial) carcinoma involving the prostate	Generally not detectable ; stromal lesions hypoechoic in 71%	Must be biopsy-confirmed : bacillus Calmette–Guérin (BCG) granulomas are also hypoechoic
Squamous cell carcinoma	Irregular anterior mass, relatively hyperechoic	Extension from bladder or urethra is much commoner than primary prostatic SCC
Adenoid cystic / basal cell carcinoma	Multiple, evenly distributed small anechoic cysts	Rare but potentially fatal; cribriform or adenoid cystic histology
Sarcoma	Irregular hypoechoic mass with anechoic necrosis	Rhabdomyosarcoma is isoechoic to normal prostate
Haematological / lymphoid	Generally not visualised	Lymphocytic infiltrate on biopsy often misattributed to chronic inflammation

- **Bladder wall thickening** on prostate TRUS usually reflects **BPH-related hypertrophy**, but may suggest bladder malignancy, as may an incidental bladder mass.

PROSTATE BIOPSY: TECHNIQUES AND OUTCOMES

Indications for Prostate Biopsy

- Screening debate: Chapter 157. **Pre-PSA**: DRE suspicion → digitally directed biopsy; **PSA screening made TRUS biopsy standard of care**. TRUS without biopsy has little value for cancer.
- **Initial biopsy detects cancer in ≈30%**. **Core needle biopsy is the technique of choice**; fine-needle aspiration is obsolete.
- Screening raised **organ-confined, curable disease** but brought **overdiagnosis and overtreatment**.

- **Historic trigger: PSA >4.0 ng/mL.** The **Prostate Cancer Prevention Trial (PCPT)** showed **no safe PSA threshold at any age**: with PSA ≤4.0, 15% had cancer, and nearly 15% of those were Gleason ≥7.

SCREENING TRIALS: ERSPC VERSUS PLCO

Trial	Result on prostate cancer mortality
European Randomized Study of Screening for Prostate Cancer (ERSPC)	Net benefit
Prostate, Lung, Colorectal and Ovarian (PLCO) Cancer Screening Trial (United States)	No benefit on cancer-specific survival
Reanalysis of both using mean lead-time estimates	Both give “ compatible evidence that screening reduces prostate cancer mortality ”. The accompanying editorial: the question is no longer whether screening can do good, but whether behaviour can change so that it does more good than harm

- Organisations range from **informed decision-making** to **rejecting screening**. Weigh health, age, family history, options, wishes and growing concern over **biopsy morbidity and mortality**; online nomograms help.
- **Elevated risk**: age >50; family history and >45; African ancestry; **baseline PSA >1 ng/mL at 40 or >2 ng/mL at 60**.
- **Absolute PSA cut-offs largely abandoned** for risk stratification and PSA kinetics. **Adjuncts** (free-to-total ratio, velocity, PSAD) are **not uniformly reliable**; **genomic markers** address PSA's **poor specificity**; **inherited risk** increasingly recognised.

GUIDELINE BODIES: SCREENING AND THE DECISION TO BIOPSY

Body	Position
AUA / Society of Urologic Oncology (SUO), 2023	Shared decision-making . Baseline PSA at 45–50; regular PSA every 2–4 years at 50–69. Germline mutation, strong family history or African ancestry: start at 40–45
National Comprehensive Cancer Network (NCCN), 2023	Biopsy for abnormal DRE with raised PSA , or repeat PSA >3 ng/mL with abnormal mpMRI (Prostate Imaging Reporting and Data System [PIRADS] version 2 ≥3) . PSA velocity alone, or abnormal DRE with normal PSA, is insufficient but may be combined with other risk factors. Selective biomarkers before initial or repeat biopsy (below)
US Preventive Services Task Force (USPSTF), 2018	Age 55–69: PSA screening is an individual decision after discussing risks and benefits

NCCN BIOMARKERS FOR THE DECISION TO BIOPSY OR RE-BIOPSY

Category	Assay	Target
Serum PSA derivatives	Prostate Health Index (PHI), 4Kscore	PSA isoforms / kallikreins
Tissue	ConfirmMDx	Stromal (field-effect) hypermethylation
Urine	MiPS, ExoDx	ERG
Urine	SelectMDx	DLX1 and HOXC6
Urine	PCA3	PCA3

- **Prebiopsy mpMRI** is now widely endorsed and enables targeted sampling.

Box 155.1 Commonly Cited Indications for Transrectal Ultrasonography Alone and Transrectal Ultrasonography With Initial and Follow-Up Prostate Biopsy

- **TRUS without biopsy:** treatment-planning volumes (brachytherapy, cryotherapy, BPH therapy such as transurethral microwave thermotherapy and radiofrequency ablation); volume during hormonal downsizing before radiotherapy; fiducial or radiofrequency marker placement; **azoospermia work-up** (ejaculatory duct and SV cysts); aspiration or unroofing of cysts and **drainage of prostatic abscess**.
- **Initial biopsy:** informed-choice diagnosis in an asymptomatic man (PSA plus risk factors); **very suspicious DRE or nodule (5–30% cancer risk)**; symptoms suggesting cancer; findings suggesting metastasis (bone lesions, adenopathy); cancer found at transurethral resection done for presumed benign disease.
- **Repeat biopsy:** rising or persistently raised PSA; **atypical intraductal proliferation (50%)**; **atypical small acinar proliferation (ASAP; 50% cancer risk)**; positive urinary PCA3 or newer genomic tests (e.g. ConfirmMDx) with persistent suspicion; suspicious MRI lesion with persistent PSA elevation; local versus systemic recurrence after ablative therapy.
- **Follow-up biopsy:** active surveillance protocol.

Modified from Heidenreich et al. (2014); Thomsen et al. (2014); Gomella and Amirian (2015).

Contraindications to Prostate Biopsy

- **Contraindications:** significant coagulopathy; severe immunosuppression; acute urinary tract infection (UTI) or prostatitis.
- **Painful anorectal disease or anal stenosis** → general or regional anaesthesia.
- **No anus** (after abdominoperineal resection) → **in-bore MRI biopsy**, or **transperineal biopsy with an end-fire probe on the perineum** (theatre under general, or local/regional by an experienced sonographer).
Limitation: the prostate lies centimetres away, so **definition is poor**, especially base and SVs.

Preparing Patients for Biopsy

- **Informed consent; stop herbal supplements** (undeclared agents); **a little urine in the bladder** shows the prostatovesical junction.

ANTITHROMBOTIC MANAGEMENT BEFORE BIOPSY

Agent	Action
Low-dose aspirin	Continue
Warfarin; clopidogrel (an antiplatelet agent)	Stop 5–7 days before
Direct oral anticoagulants: apixaban, dabigatran, rivaroxaban	Stop 2–5 days before. Rivaroxaban: stopping may raise stroke risk → bridge with short-acting heparin
Warfarin or coagulopathy, low thromboembolic risk	Do not biopsy until international normalised ratio (INR) <1.5
Warfarin, high thromboembolic risk (e.g. mechanical valve)	Bridge with unfractionated or low-molecular-weight heparin

Antibiotic Prophylaxis

- Rising complications, **mostly infective admissions**, renewed the focus on prophylaxis.
- **Transrectal biopsy is Class III (contaminated):** unlike other lower-tract procedures, **prophylaxis for every patient, regardless of risk factors**.
- **AUA 2019 best practice:** fluoroquinolones; first- or second-generation cephalosporin ± aminoglycoside; third-generation cephalosporin (**evidence level Ib**). **Intramuscular acceptable for all; oral only for quinolones**. Give **≥1 hour before; ≤24 hours**.

- **Transient bacteraemia is common**, usually silent. **Endocarditis or prosthetic joint, pacemaker or defibrillator risk**: IV ampicillin (**vancomycin if penicillin-allergic**) + gentamicin → oral fluoroquinolone for 2–3 days.
- **Food and Drug Administration warning (2016)**: tendinitis and rupture, peripheral neuropathy, central nervous system effects, myasthenia gravis → AUA caps fluoroquinolone prophylaxis at **24 hours**.
- **Cochrane review (2011)**: antibiotics cut bacteriuria, bacteraemia, fever, UTI and admission versus placebo; **longer or multi-dose courses no better than short or single-dose**.

TABLE 155.1 AMERICAN UROLOGICAL ASSOCIATION 2019 BEST-PRACTICE POLICY RECOMMENDED ANTIBIOTICS FOR ROUTINE PROSTATE BIOPSY*

Class	Agents and dosing	Route pattern
Fluoroquinolones†	Levofloxacin 500 mg single dose ; ciprofloxacin 500 mg q12h; ofloxacin 400 mg q12h	Oral
Aminoglycosides‡	Gentamicin 5 mg/kg ; tobramycin 5 mg/kg; amikacin 15 mg/kg , each a single dose	IV single dose
First-generation cephalosporins	Cephalexin 500 mg q6h; cephadrine 500 mg q6h; cefadroxil 500 mg q12h; cefazolin 1 g IV q8h	Oral, except cefazolin
Second-generation cephalosporins	Cefaclor 500 mg q8h; cefprozil 500 mg q12h; cefuroxime 500 mg q12h; cefoxitin 1–2 g IV q6h	Oral, except cefoxitin
Third-generation cephalosporins	Ceftizoxime 1 g q8h; ceftazidime 1 g q12h; ceftriaxone 1–2 g single dose ; cefotaxime 1 g q8h	All IV
Alternatives	Aztreonam 1–2 g IV q8h ; trimethoprim-sulfamethoxazole 1 double-strength tablet q12h	IV / oral

*Duration ≤24 hours (level of evidence Ib). †Fluoroquinolones: increased risk of tendinitis and tendon rupture. ‡Aztreonam can replace aminoglycosides in renal insufficiency. Modified from the AUA Best Practice Policy Statement on Urologic Surgery Antimicrobial Prophylaxis.

- **What the table tests**: the **single-dose agents** (levofloxacin, the aminoglycosides, ceftriaxone); the **aztreonam-for-aminoglycoside swap in renal impairment**; and fluoroquinolone tendon toxicity.
- **Trap**: the table lists oral cephalosporins and trimethoprim-sulfamethoxazole, yet the **2019 statement recommends the oral route only for quinolones**. The table is the general AUA surgical prophylaxis list, not route advice specific to transrectal biopsy.
- **Rectal swab-directed prophylaxis**: a resistant organism on swab **does not reliably predict infection**; **cost-effectiveness unproven** against empirical therapy guided by local susceptibility. A **risk-adjusted strategy** (resistance monitoring, combined oral + intramuscular agents, selective swabbing of high-risk men) works prospectively.
- **Antimicrobial exposure within 6 months is the most consistent risk factor** → **augmented (>1 agent) or targeted prophylaxis**.

Table 155.2 Risk Factors for Prostate Biopsy Infection

- **Medical history**: diabetes; significant comorbidity (**Charlson score >1**); immunosuppression (steroids, chemotherapy, HIV).
- **Urological history**: previous UTI or prostatitis; previous biopsy infection; **greater cumulative number of biopsies**.
- **Exposure to antibiotic resistance**: **antibiotics in the last 6 months**; recent international travel (endemic resistance in **India and South-East Asia**; antibiotics taken to prevent traveller's diarrhoea); **health-care worker**.
- **Colonisation**: fluoroquinolone-resistant *Escherichia coli* on rectal culture.

From AUA: Prevention and Treatment of the More Common Complications Related to Prostate Biopsy Update.

- **Pattern:** each factor marks either **a host that cannot contain bacteraemia or a rectum colonised by resistant organisms**; the responses are augmented or targeted prophylaxis, or avoiding the rectum (transperineal).

Concept — *The transrectal needle inoculates rectal flora into the prostate with every core. Prophylaxis therefore has to be **universal**, and its failures are driven by **fluoroquinolone resistance**.*

Cleansing Enema

- Self-administered; **improves the acoustic window**; **no proven effect on infection**, so some omit it.

Analgesia

- **TRUS-guided periprostatic block:** 1–2% lidocaine, 7-inch 22-gauge spinal needle via the biopsy channel. Variants: 5 mL each side at the hyperechoic SV–prostate junction fat pad (authors'); 10 mL from SV junction along the lateral gland, base to apex; **intraprostatic injection augments. Saturation: up to 22 mL of 1%**; avoid intravascular injection.
- **Topical intrarectal anaesthetic is inferior to periprostatic block**; useful when probe insertion is tender.

Patient Positioning

- **Left lateral decubitus, knees and hips at 90°**, arm board, pillow between knees, **buttocks flush with the table edge**; right lateral or lithotomy if needed.
- **Lithotomy preferred for:** transperineal biopsy, brachytherapy planning, **Doppler and contrast studies (gravity distorts vascular patterns)**, fiducial markers.

Transrectal Prostate Biopsy Techniques

- **Sequence:** DRE (nodularity, anal pathology) → volume → transverse and sagittal survey **from base to apex**, noting lesions (hypo- or hyperechoic, calcification, contour, cysts) → biopsy.
- **Spring-driven 18-gauge gun** through the needle guide; path best seen **sagittally** on a ruled puncture line.
- **Mechanics:** throw **15–25 mm**; core **1.0–1.5 cm**; tip runs **0.5 cm beyond the sample**, so advancing past the capsule samples **anterior tissue and misses the PZ. Technique: indent the capsule → withdraw a few millimetres → fire.**
- **Less bleeding and pain:** never reposition with the needle on the rectal wall; **compress the mucosa with the probe** (like pulling skin taut for phlebotomy).
- **Specimens:** 10% formalin. AUA white paper: **site-specific labelling not shown to change management; ≤2 cores per jar** (overpacking lowers detection).

Sextant Biopsy

- **Hodge (1989): base, mid and apex bilaterally, parasagittal**; a major advance on digitally and lesion-directed biopsy.
- **Flaw:** parasagittal cores took **much TZ, little PZ**, while most cancers arise in the **posterolateral PZ** (McNeal); hence sextant false negatives.

Extended-Core Biopsy Techniques

- **Laterally directed cores** came first and consistently raise detection; **six cores are now inadequate.**
- **Extended 12-core** (adds **apical and far-lateral** cores): maximal detection, fewer repeat biopsies, fewer insignificant diagnoses; **AUA-endorsed.**
- **6 → 12 cores raises detection; 12 → 18–21 (saturation) initially does not.** Yet initial saturation meant **less cancer on later biopsy**, and later cancers were mostly insignificant. Saturation is mainly **for prior negatives.**

- **TZ and SVs not routinely sampled** (low initial yield). **TZ/anterior cores**: persistent PSA after negative biopsy, or initially if **gland >50 mL**; MRI now finds anterior tumours. **SV biopsy**: palpable abnormality; some add **PSA >30 ng/mL** or pre-brachytherapy.

FIGURE 155.5 SYSTEMATIC BIOPSY SCHEMES

Scheme	Cores	Source	Status
Sextant	6	Hodge and associates, 1989	Inadequate today
10-core	10	Presti et al., 2000	Extended
12-core (double sextant)	12	Bjurlin et al., 2013	Currently recommended; AUA-endorsed
13-core five-region	13	Eskew et al., 1997	Extended
Saturation	>12 (18–21 or more)	Not applicable	Repeat setting

Repeat and Saturation Prostate Biopsy

- **Negative biopsies with persisting concern** is common; **yield falls with each successive biopsy**.
- **ERSPC**: after initial sextant, yield on biopsies 2–4 was **22% → 10% → 5% → 4%**. **Positive predictive value (PPV) held steady** for biopsy-naïve men across rounds but **fell for previously negative men**; **repeat biopsies were a quarter of all biopsies yet found under a tenth of aggressive cancers**.
- **Saturation (>12 cores) after negatives**: early series improved detection, but **anaesthetic needs mean hospital**, and a later large prospective study found **no significant gain**.
- **Second biopsy justified** if concern persists; **third and fourth only for high suspicion or poor prognostic features**; **target the apex**. Now add genomic testing of prior tissue and mpMRI.

Transurethral Prostate Biopsy

- Once used for TZ cancer or after negative TRUS. **Solitary TZ cancers: <5%**. Now largely obsolete: **TRUS under local samples the TZ**, **MRI (especially fusion) finds anterior cancers**, and **transperineal biopsy** reaches them.

Concept — Biopsy strategy evolved by **following where cancer actually lies**, from palpable nodules to the posterolateral PZ and then to anterior and apical tumours. Each gain came from **better geometry, not more cores**: 12 beats 6, but 21 does not beat 12.

Risks and Complications of Transrectal Prostate Biopsy

- **Serious complications needing admission: <2%**, but in a large population series **30-day admissions rose (1.0% → 4.1%)**, mostly infective.
- **PLCO and ERSPC: no excess mortality**; infection **<1%**. Fear alone should not deter healthy men with long life expectancy.

COMPLICATIONS OF TRANSRECTAL BIOPSY

Complication	Reported rate	Key management
Haematuria	23–63%; clot retention 0.7%	Self-limiting. Clot retention → catheter, manual washout; severe → continuous bladder irrigation
Rectal bleeding	2.1–21.7%	Pressure with probe or finger; escalation below
Haematospermia	9.8–50.4%	Counsel beforehand. Lasts about a month , over several ejaculations; no predictor of duration
Infection (any)	0.1–7.0%; UTI 2–6%	Mostly UTI or low-grade fever; oral or IV antibiotics
Sepsis	0.3–3.1%	Carbapenem until cultures return

Complication	Reported rate	Key management
Hospitalisation for infection	0.6–4.1%	Up to 2 weeks of antibiotics
Vasovagal reaction (moderate or severe)	1.4–5.3%	May end the procedure; Trendelenburg + IV fluids
Acute urinary retention	0.2–0.4%	Temporary catheter; risk with large glands, high International Prostate Symptom Score (IPSS)
Erectile dysfunction	Incompletely characterised	Transient; anxiety contributes

- **“Most common complication” has two answers: bleeding overall** (even with normal coagulation) and **infection among serious complications** and admissions: **frequency versus severity**.
- **Two admission figures:** <2% is all serious complications; **0.6–4.1%** is contemporary infective admission, its top end from the **later, higher-resistance era**.

Postbiopsy Infections

- **Fluoroquinolone-resistant *E. coli*** dominates, bred by fluoroquinolone use in health care and the **food supply**. Admissions and fatal sepsis are rising, but sepsis mortality is **no higher than for similar infections**.
- **Sepsis:** systemic inflammatory response to infection (fever, hypothermia, tachypnoea, tachycardia, confusion, hypotension). **Septic shock:** hypotension **despite adequate fluid resuscitation**. **Any post-biopsy fever → assess for sepsis**.
- **Prevention:** swab-targeted prophylaxis has **some utility**; **no bowel preparation helps**; **transperineal** is promising.
- **Sensitivities:** resistant to **fluoroquinolones, trimethoprim-sulfamethoxazole, ampicillin**; susceptible to **second- and third-generation cephalosporins, amikacin, carbapenems**. Risk factors as Table 155.2, plus **number of prior biopsies** (active surveillance).
- **Treatment:** **no empirical fluoroquinolone → carbapenem for sepsis** until cultures → **up to 2 weeks** if admitted.

Bleeding

- **Intermediate bleeding-risk procedure:** hold antithrombotics and herbal remedies by class; **low-dose aspirin** may continue.
- **Persistent brisk rectal bleeding, escalation:** direct pressure (probe or finger) → **tamponade with an inflated probe sheath or Foley balloon** → anoscopy or colonoscopy with **adrenaline and polidocanol** or other sclerosants → **angiographic embolisation** → transrectal exploration and suturing.

Other Complications

- **Vasovagal reaction** stems from anxiety and probe discomfort. Any biopsy may raise the IPSS, but only **transiently**.
- **Erectile dysfunction:** first blamed on neurovascular bundle injury, but data conflict. It is worse in men **diagnosed with cancer**, which implicates **anxiety**.

Concept — Transrectal biopsy's complications split by **route**. **Bleeding is frequent but trivial; infection is rare but dangerous**, and the danger tracks antibiotic resistance. This asymmetry drives both the prophylaxis rules and the move to the perineum.

ADVANCED AND INVESTIGATIONAL TECHNIQUES FOR PROSTATE BIOPSY

Colour and Power Doppler Transrectal Ultrasonography

COLOUR VERSUS POWER DOPPLER

	Colour Doppler	Power Doppler (enhanced colour, colour amplitude imaging, colour angiography)
Principle	Frequency shift	Amplitude shift
Dependence	Velocity- and direction-dependent	Velocity- and direction-independent
Display	By convention red toward , blue away from transducer; not artery- or vein-specific	Flow presence only
Strength	High specificity, low sensitivity for cancer	Detects slower flow , less angle-dependent, so better for tumour neovascularity
Cancer detection	Neither proven superior ; neither accurate enough to replace systematic biopsy	

- Doppler beats grey scale alone, but **many cancers stay invisible to every sonographic mode**.
- **Rationale**: angiogenesis raises **microvessel density**, which tracks **metastasis, stage and disease-specific survival** and is higher in cancer than surrounding tissue.
- **Prognosis**: colour flow in the dominant tumour → **markedly more PSA recurrence after radical retropubic prostatectomy**, higher grade, more SV invasion; power Doppler signal also tracks Gleason score.
- **Unenhanced Doppler cannot see tumour microvessels (10–15 µm)**; it detects **larger feeding vessels**.
- **Microbubbles** (as in echocardiography): **1–10 µm** air or heavy gas in albumin or polymer shells, or lipid or surfactant coats; they amplify microvascular signal.
- **Contrast-enhanced TRUS (CE-TRUS)**: **higher sensitivity, specificity preserved**; targeting enhancing lesions improves detection; recommended for routine care by a European multicentre trial; **3D** enhanced power Doppler adds accuracy (Fig. 155.7: lesion seen only after contrast).
- **Flash-replenishment imaging** (software): high-power flash pulses **destroy microbubbles** → low-power pulses show **replenishment** → maximum-intensity composite of the vascular architecture, **down to capillary level** → real-time targeted biopsy. **Targeted cores are more often positive than systematic cores**.
- **Elastography** may outperform Doppler: **stiff tissue suggests malignancy** (Fig. 155.8: stiff right-base focus, Gleason 4+4=8). **Early: manual compression strain, operator-dependent**; **newer: beam compression or shear-wave velocity, operator-independent**. Shear-wave **stiffness rises with Gleason score**, but targeted biopsy **often missed cancer** in a large series, so it **complements, not replaces, systematic biopsy**.

Other Newer and Investigational Imaging and Biopsy Techniques

- **Prostate HistoScanning**: proprietary **backscatter** tissue analysis; **unreliable for cancer** and **inferior to mpMRI**.

Multiparametric Prostate Ultrasound

- Ultrasound analogue of mpMRI: **grey scale + vascularity (Doppler, contrast) + stiffness**; alternative or complement to MRI.

Micro-Ultrasound

- **29 MHz** (versus conventional 7–9 MHz) → resolution **70 µm** (versus 150–300 µm), close to **histological architecture**.
- **Prostate Risk Identification using Micro-Ultrasound (PRI-MUS)**: validated lesion scoring analogous to PIRADS.

- Multicentre registry: micro-ultrasound (MUS) targeting **matched or slightly beat mpMRI** for sensitivity and negative predictive value (NPV); **specificity and PPV low for both**. **No IV contrast**: key for **renal insufficiency**. Needs validation.

ADVANCED ULTRASOUND MODALITIES COMPARED

Modality	What it detects	Status
Colour / power Doppler	Feeding-vessel flow	Adjunct only
CE-TRUS, flash replenishment	Capillary-level neovascularity	Improves targeting
Elastography	Stiffness	Complement only
HistoScanning	Backscatter signature	Unreliable
Multiparametric ultrasound	All of the above	Emerging
Micro-ultrasound	Near-histological architecture	≈ mpMRI; no contrast

Concept — Every ultrasound advance tries to see one of **two biological signatures of cancer: neovascularity** (Doppler, contrast, flash replenishment) or **stiffness and architecture** (elastography, micro-ultrasound). None yet removes the need for systematic sampling.

Multiparametric Prostate Magnetic Resonance Imaging

- **Dynamic contrast enhancement (DCE) and diffusion-weighted imaging (DWI)** improve detection; AUA policy statement (2017). **Tissue still required**. Real-time MRI biopsy is hard, hence three targeting techniques: **in-bore, cognitive, fusion**.
- **PIRADS: v1 European Society of Urogenital Radiology (ESUR), 2012 → v2 American College of Radiology (ACR) + ESUR + AdMeTech, 2015** (published 2016), adding technical standards and a **reporting lexicon**.
- **Likert 1–5 for clinically significant cancer: Gleason ≥ 7 and/or $>0.5 \text{ cm}^3$ and/or EPE. PZ and TZ/CZ scored separately.** T2-weighted imaging (T2WI) and DWI scored 1–5; **DCE only positive or negative**.
- **Lesion $>1.5 \text{ cm}$ on T2WI, or highly suspicious for EPE → automatically PIRADS 5.**

Box 155.2 PIRADS v2 Categories

- **1 very low:** clinically significant cancer highly unlikely. **2 low:** unlikely. **3 intermediate:** equivocal. **4 high:** likely. **5 very high:** highly likely.

TABLES 155.3 AND 155.4 PIRADS V2 SCHEME FOR PERIPHERAL ZONE AND TRANSITION ZONE LESION ASSESSMENT CATEGORIES

	Peripheral zone (Table 155.3)	Transition zone (Table 155.4)
Dominant sequence	DWI	T2WI
Dominant score 1, 2, 4, 5	PIRADS = DWI score (T2WI and DCE any)	PIRADS = T2WI score (DWI and DCE any)
Dominant score 3: tie-breaker	DCE: negative → 3; positive → 4	DWI: $\leq 4 \rightarrow 3$; DWI 5 → 4
Role of DCE	Upgrades 3 → 4 only	No role

"Any" = any score 1–5. From Weinreb JC, Barentsz JO, Choyke PL, et al.: PIRADS prostate imaging—reporting and data system: 2015, version 2, Eur Urol 69:16–40, 2016.

- **What the tables test:** **DWI rules the PZ, T2WI rules the TZ**, and each zone has a single tie-breaker used **only at score 3**. DCE is a secondary sequence that **never moves a lesion by more than one category** and is irrelevant in the TZ.
- **Trap:** assuming DCE can create or downgrade a PIRADS 4–5, or that a TZ lesion is upgraded by enhancement.
- **Validation (German, Dutch):** cancer in **≈1/3 of PIRADS 3, over half of PIRADS 4, ≈90% of PIRADS 5**; clinically significant in roughly half of each. mpMRI can therefore **limit biopsies and cores**.

LANDMARK MRI-PATHWAY TRIALS

Trial	Design and population	Comparator and definition	Key results	Verdict
PROMIS (2017)	Paired-cohort, multicentre confirmatory; 576 men; all had mpMRI, 5-mm transperineal template mapping biopsy (reference) and 12-core TRUS biopsy	Clinically significant cancer = Grade Group ≥ 3 or maximum cancer core length ≥ 6 mm	mpMRI vs TRUS: sensitivity 93% vs 48%; NPV 89% vs 74% . Triage could avoid biopsy in 27% , cut insignificant diagnoses by 5% ; targeting could detect 18% more significant cancers	mpMRI is a better triage test than TRUS biopsy
PRECISION (2018)	Multicentre randomised non-inferiority ; 500 men	mpMRI → targeted biopsy of PIRADS ≥ 3 vs standard 12-core TRUS biopsy	Clinically significant detection 38% vs 26%; 28% avoided biopsy (no target)	MRI pathway non-inferior, in fact superior

- **Adding systematic cores** to targeted: controversial; a large prospective trial showed a **small absolute gain**.
- **mpMRI misses a meaningful minority of significant cancers** even at expert centres (prostatectomy reference); most misses are 3+4, but a third are **primary pattern 4**. Causes: MRI-invisible lesions, reader discordance, segmentation, coregistration or biopsy error, **small lesion in a large gland**, variant histology, low PIRADS.

Transrectal Targeting**MRI-GUIDED TARGETING TECHNIQUES**

	In-bore MRI	Cognitive	MRI/ultrasound software fusion
Method	Patient prone in gantry , needle guide in rectum. Multisequence localisation, then short True Fast Imaging (TRUFI) sequences guide the needle	Operator reviews MRI, then aims mentally during standard TRUS biopsy (memory, 3D reasoning, pattern recognition)	Software coregisters premarked MRI with real-time TRUS into a 3D model, with elastic deformation to correct probe distortion
Equipment	Specialised MRI kit	None extra	Fusion platform
Drawbacks	Time, cost, logistics , limited space	Cannot record or confirm needle position ; higher interoperator variability; poor with no landmarks (small lesion in large gland) or lesion invisible on TRUS	Cost; coregistration error
Mitigation	Not applicable	2–4 cores per target + systematic biopsy	Not applicable
Accuracy	Meta-analysis (2022): no significant difference among the three in significant or insignificant cancer detection		

FUSION PLATFORMS: URONAV VERSUS ARTEMIS

UroNav (Philips/Invivo)	Artemis with ProFuse (Eigen)
Freehand standard probe; electromagnetic tracking by an external field generator	Mechanical arm directs the biopsy
Studied at the National Cancer Institute (NCI) since 2004; largest published experience	The other most-used US system

- Acquisition differs (freehand sweep vs mechanical stabilisation); all use a **standard spring-loaded needle** (Fig. 155.9: live ultrasound above, MRI below, coregistered target and needle tract).

- **Fusion results: PIRADS 5 nearly always positive;** targeted cores **≈3× likelier** to find intermediate- or high-risk disease; suspicion tracks grade (high sensitivity and NPV for Gleason ≥8); frequent **upgrading** versus 12-core.
- **Targeting finds high-grade and misses low-grade cancer:** a feature, as it curbs overdiagnosis.
- **Indications:** raised PSA (initial); persistent PSA after negative biopsy; **active surveillance** (very low, low, favourable intermediate); **focal therapy planning**.

GUIDELINE BODIES: MRI-TARGETED BIOPSY

Body	Recommendation
AUA + Society of Abdominal Radiology (SAR), 2016	mpMRI and targeted biopsy of PIRADS ≥3 after negative prior biopsy with persistently raised PSA
NCCN Early Detection, 2023	Incorporated the same strategy
AUA/SUO 2023 (Early Detection Part II)	Targeted biopsy whenever biopsy is indicated and a PIRADS ≥3 lesion exists; 2–3 cores per target ; synchronous systematic biopsy allowed

Concept — MRI changed biopsy from **sampling a template to sampling a target**. The result is fewer biopsies, more significant and fewer insignificant cancers, but **tissue is still mandatory** and MRI-invisible disease keeps systematic cores relevant.

Transperineal Prostate Biopsy

- Transperineal adoption may overtake the transrectal gold standard: **fewer infections, stewardship by omitting antibiotics, apex and anterior access**, better local analgesia. Slogan: “TREXIT”.
- **Uptake:** nearly **doubled in the United Kingdom**; **over half of biopsies in Australia and New Zealand**, with geographic and socioeconomic variation.
- **European Association of Urology (EAU) 2023:** transperineal preferred over transrectal for systematic biopsy to detect clinically significant cancer (**strong** rating).

TRANSRECTAL VERSUS TRANSPERINEAL BIOPSY

	Transrectal	Transperineal
Position	Left lateral decubitus	Lithotomy
Needle passes through	Rectal wall (contaminated)	Sterilisable perineal skin
Prophylaxis	Mandatory for all	Skin-flora only, or none
Infection	Higher; fluoroquinolone resistance-driven	Lower (meta-analysis of randomised trials)
Rectal bleeding	Common	Eliminated
Apex and anterior / TZ access	Needs excessive probe torque	Ready access ; full core length through PZ
Template	Single established 12-core	Several; none universally accepted
Guideline standing	Chapter's gold standard	EAU 2023 preferred ; randomised trials under way

Techniques

- **History:** the transperineal route **preceded** the transrectal one. **Barringer (1922)** perineal needle biopsy, with mixed success → open perineal cut-down → transperineal needle aspiration under rectal finger guidance → modern TRUS-guided biopsy.
- **Positioning.** **Lithotomy, biplanar probe in rectum**, needle through perineum; **scrotum taped up and forward** to put the perineum under tension, removing laxity so landmarks and targeting are reproducible.

Access

TRANSPERINEAL ACCESS METHODS

Method	Features	Trade-off
Stepper + brachytherapy grid (5-mm increments)	Mapping or targeting	Multiple skin punctures ; probe's freedom of movement limited
Freehand needle-guide kit (PrecisionPoint most widely available)	One midline puncture (fan technique) or several bilateral; cannula fixed to probe keeps needle in view	Device cost
Two preplaced 14-gauge cannulas , decoupled from probe	Most versatile, least costly ; most freedom to angle needle	Needs sonographic skill to keep the needle in view. Office-ready after a 4–6-week learning curve in theatre; 12–14-minute procedure

Analgesia

- Formerly theatre-based (still reasonable for **many-puncture mapping**); now **freehand office biopsy under local is strongly supported**, with very rare abandonment.
- **Anaesthetise**: perineal skin, deep transverse perineal muscle, levator ani, capsule (**pudendal and perineal nerves, pelvic plexus branches**). Options include **pudendal block**.
- **Periapical triangle block** (commonest). Boundaries: **lateral, medial border of levator ani; medial, rhabdosphincter; posterior, external anal sphincter; superior, prostatic apex**.
- **Sequence**: skin and tract **1–2 mL 1% lidocaine** via 23-gauge needle → **5 mL into the periapical triangle** via 20-gauge 7-inch spinal needle → **3 mL along the capsule** (periprostic) → each step under ultrasound, watching the hydrodissection.
- **Tolerability**: peak visual analogue scale <4. Most uncomfortable to least: **anaesthetic injection → biopsy → probe insertion**.

Antibiotic Prophylaxis

- **Skin prepared with povidone-iodine or chlorhexidine**: no rectal flora is carried in.
- **Meta-analysis of randomised trials**: **significantly less infection** than transrectal; admissions similar; **low certainty**.
- Hence **skin-flora prophylaxis only**, and increasingly **none**: antibiotics **did not change the very low sepsis or infection rates** (adjusted for cores, sepsis unchanged, infection slightly higher without).

Templates

- The metric is **clinically significant cancer detection**; transperineal access reaches every region, including **apex and anterior/TZ**.

TRANSPERINEAL SYSTEMATIC TEMPLATES

Template	Division	Cores	Distinguishing feature
Original Barzell (2003)	Octants : anterior/posterior × apex/base, each side	4–8 per octant	Saturation / mapping
Modified Barzell	Quadrants (R/L apex, R/L base) × 5 sectors: parasagittal anterior and posterior, medial anterior and posterior, lateral (includes the PZ horn wrapping anterolaterally)	20	Basis of MUSIC

Template	Division	Cores	Distinguishing feature
Ginsburg	Per side: anterior (TZ > PZ), mid (apex, PZ > TZ), posterior (PZ); base (PZ) sector added if >50 g	4 per sector, medial to lateral, one jar per sector	Fewer sectors
Michigan Urological Surgery Improvement Collaborative (MUSIC)	Based on modified Barzell; PZ-focused , minimal but retained TZ sampling	Parity with 12-core TRUS	Non-inferior for overall and significant cancer. Omits midline cores → less urethral trauma, pain and haematuria. The authors' preferred template for the office
12-core; 10-sector	Also among the better-studied templates; none is universally accepted		

Cancer Detection Rate of Transperineal Prostate Biopsy

- **At least non-inferior** to transrectal. Evidence: sextant TRUS vs six transperineal fan cores (PSA >4): **comparable, slightly favouring transperineal** → randomised 6 vs 12 transperineal cores: **12 superior** (mirroring TRUS) → randomised transperineal vs TRUS, 12 cores (8 PZ + 4 TZ), spinal, PSA 4–20: **no difference** → meta-analysis under local: **comparable to historical TRUS**, varying by technique (targeted, freehand, PrecisionPoint, coaxial).
- **Anterior-only cancers: 6–20%**, which the transperineal route is **uniquely placed to sample**, as it is the apex.

Summary of Advantages of Transperineal Prostate Biopsy

- **All segments accessible, minimal learning curve**, office-based, and potentially **antibiotic-free and device-free**. Randomised trials are testing whether it should become the new gold standard.
- **Future**: genomic markers to select for initial and repeat biopsy; refined Doppler, contrast, elastography, multi- and **biparametric MRI**, **transperineal TRUS/MRI fusion**; **PSMA-PET-CT/MRI** for risk stratification and targeting.
- **Until trials report, systematic grey-scale TRUS 12-core biopsy ± targeting remains the gold standard** in this chapter, although **EAU 2023 already prefers transperineal**.

Concept — The transperineal route wins on **anatomy and microbiology at once**. The needle enters through **sterilisable skin rather than colonised rectum** and runs **along the gland's long axis**, reaching the apex and anterior zone. Its only weakness, the lack of a single agreed template, is organisational rather than biological.

KEY POINTS: PROSTATE BIOPSY

- **Imaging alone cannot diagnose cancer**; tissue is required. **Local anaesthesia** is routine.
- Transrectal: **prophylaxis ≤24 hours**; systematic biopsy **≥12 cores** (AUA).
- **Prebiopsy mpMRI** whenever possible; **fusion or cognitive targeting** improves accuracy.
- **Transperineal under local, with or without antibiotics**, is an emerging alternative. Advanced ultrasound and **PSMA imaging** may improve detection.